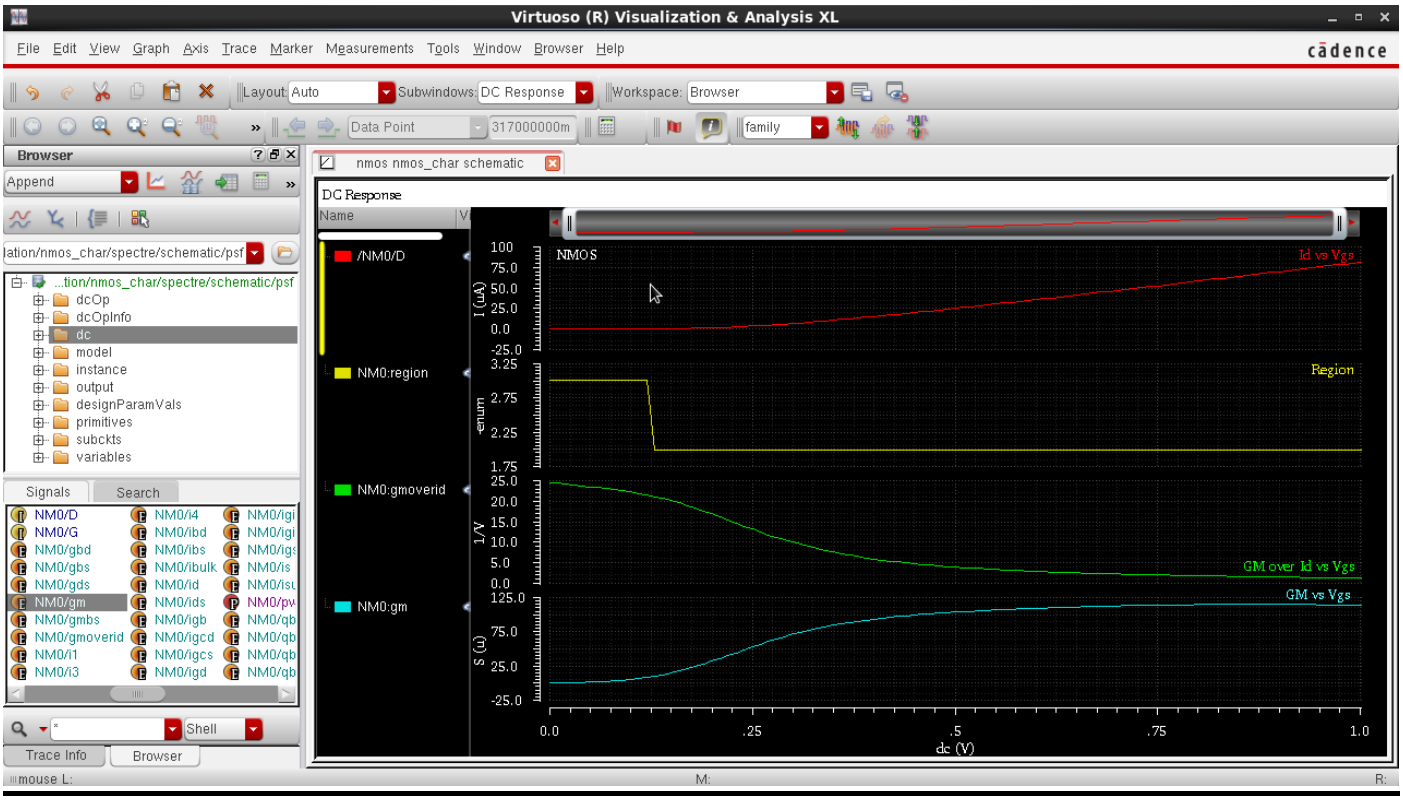
Assignment - 02

NMOS



- **Region of Operations:**
  - At low V_{gs}, the NMOS remains in the cutoff region because $V_{gs} < V_{th}$, so no inversion channel is formed.
  - When V_{gs}exceeds the threshold voltage, the transistor enters the saturation region under the applied bias and starts conducting.

- **g_m over I_d v/s Vgs:**
  - At low V_{gs}, a small drain current provides relatively high transconductance efficiency, resulting in a high g_m over I_d.
  - As V_{gs} increases, the drain current grows faster than g_m, causing g_m over $I_d$to decrease.

- **g_m v/s Vgs:**
  - Initially, g_m is nearly zero because the transistor is OFF and the drain current is negligible.
  - After channel formation, the drain current becomes more sensitive to V_{gs}, so the transconductance (g_m) increases.

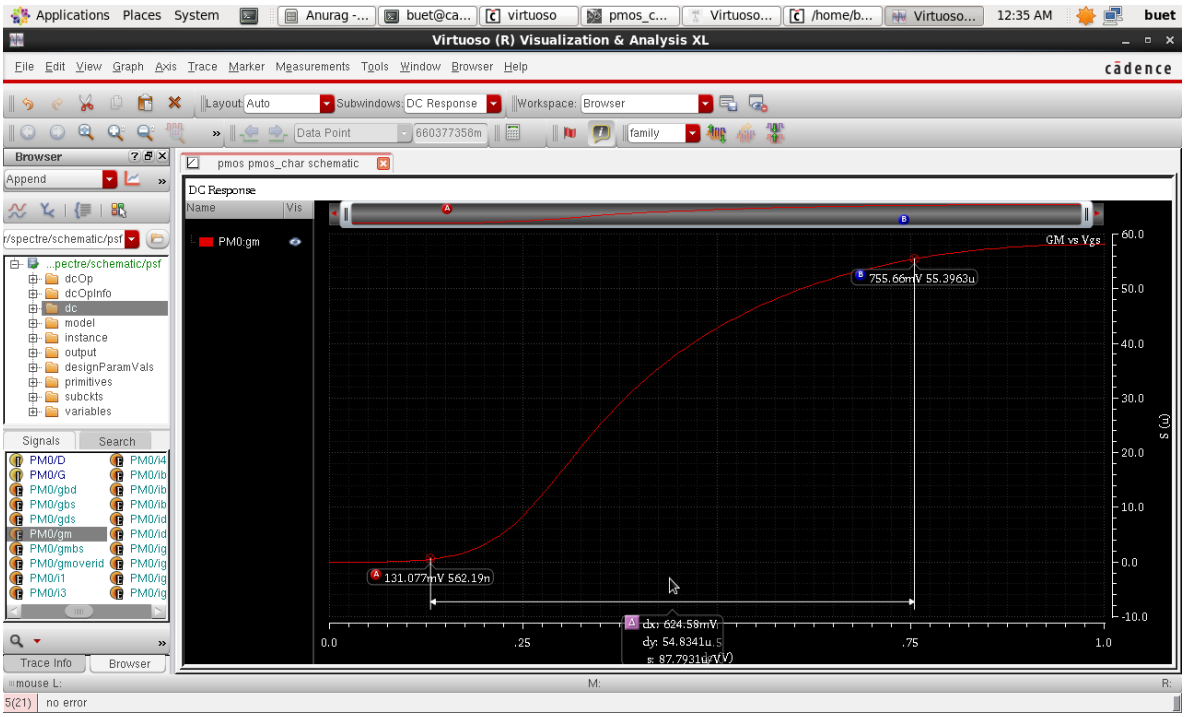
- **V_{th} v/s V_{gs} at different Temperature:**



  - The threshold voltage remains almost constant with V_{gs} because V_{th} is primarily determined by the device fabrication and temperature, not by the applied gate voltage.
  - As temperature increases, the threshold voltage decreases due to the reduction in the semiconductor surface potential and Fermi potential, so each higher-temperature curve appears at a lower V_{th}.

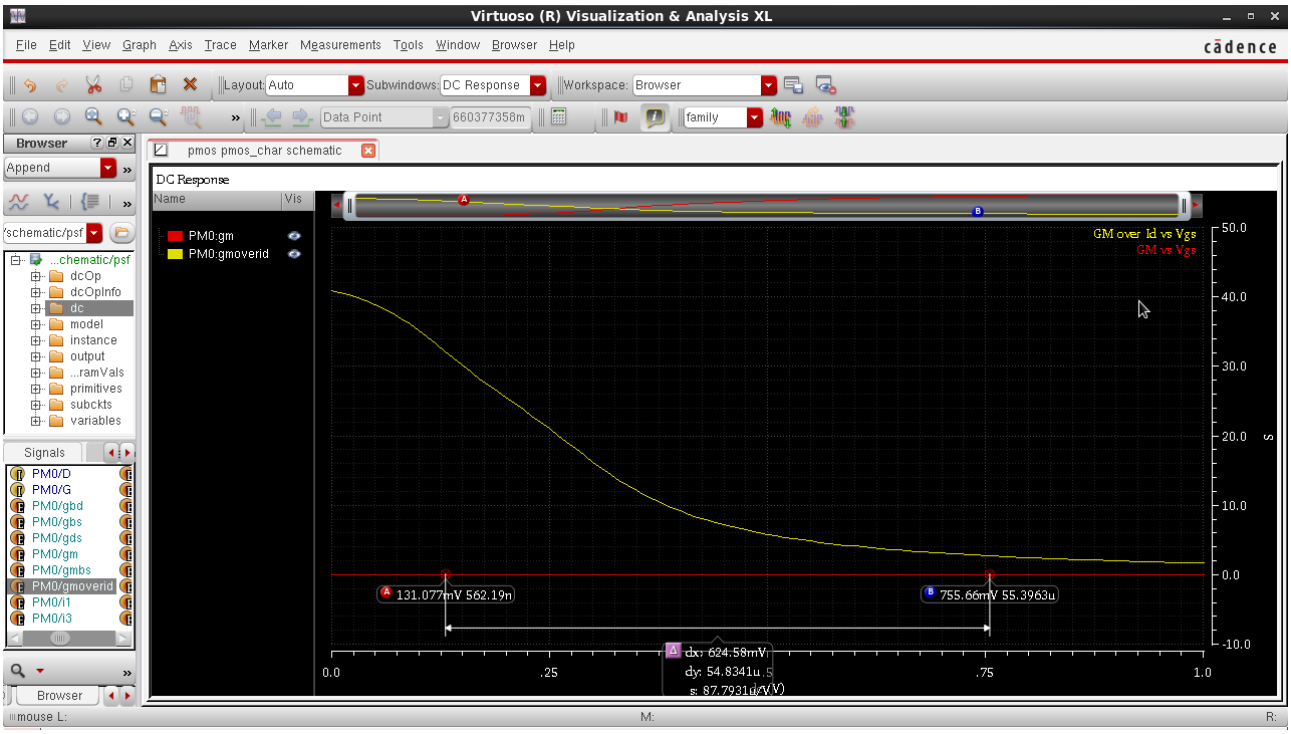
PMOS

- **g_m v/s V_{gs}:**



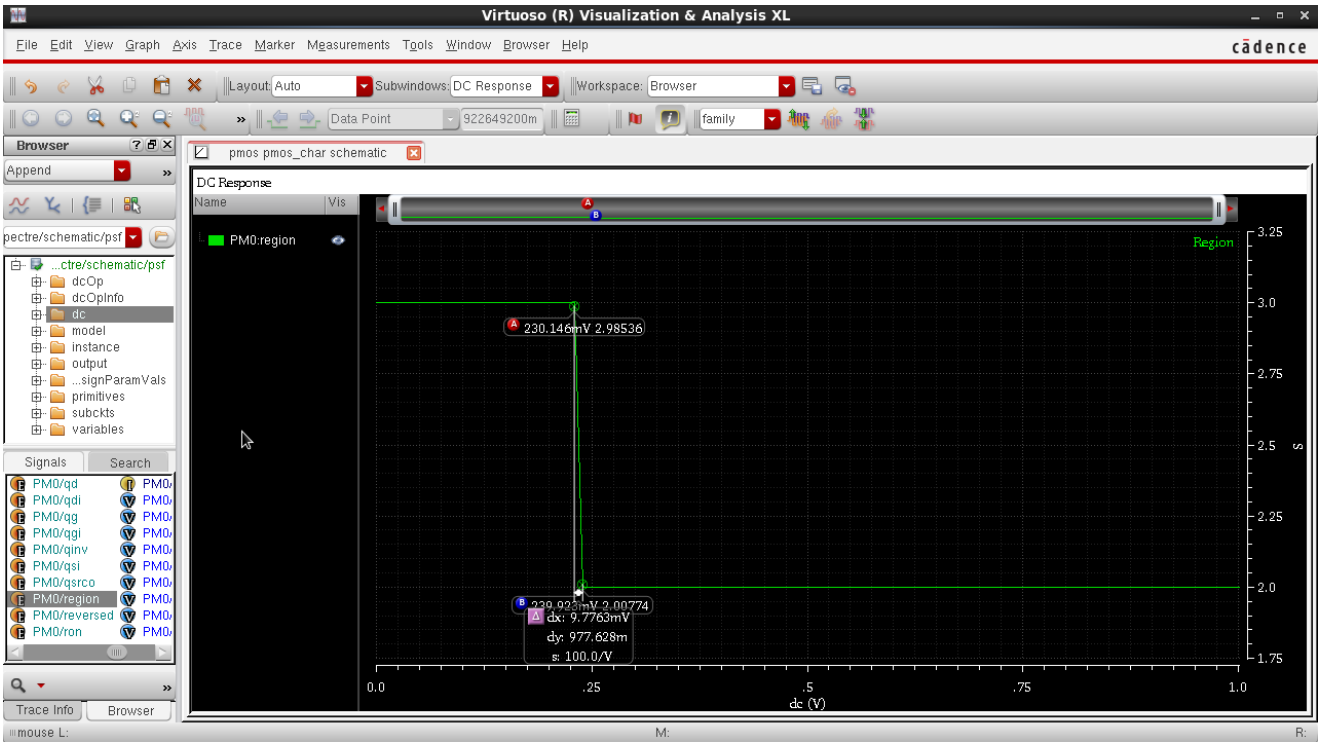
  - At low V_{gs}, the PMOS is OFF, so the transconductance (g_m) is nearly zero.
  - As | V_{gs} |increases beyond the threshold voltage, the channel becomes stronger, making the drain current more sensitive to gate voltage; therefore, g_m increases.

- **g_m over I_d v/s V_{gs}:**



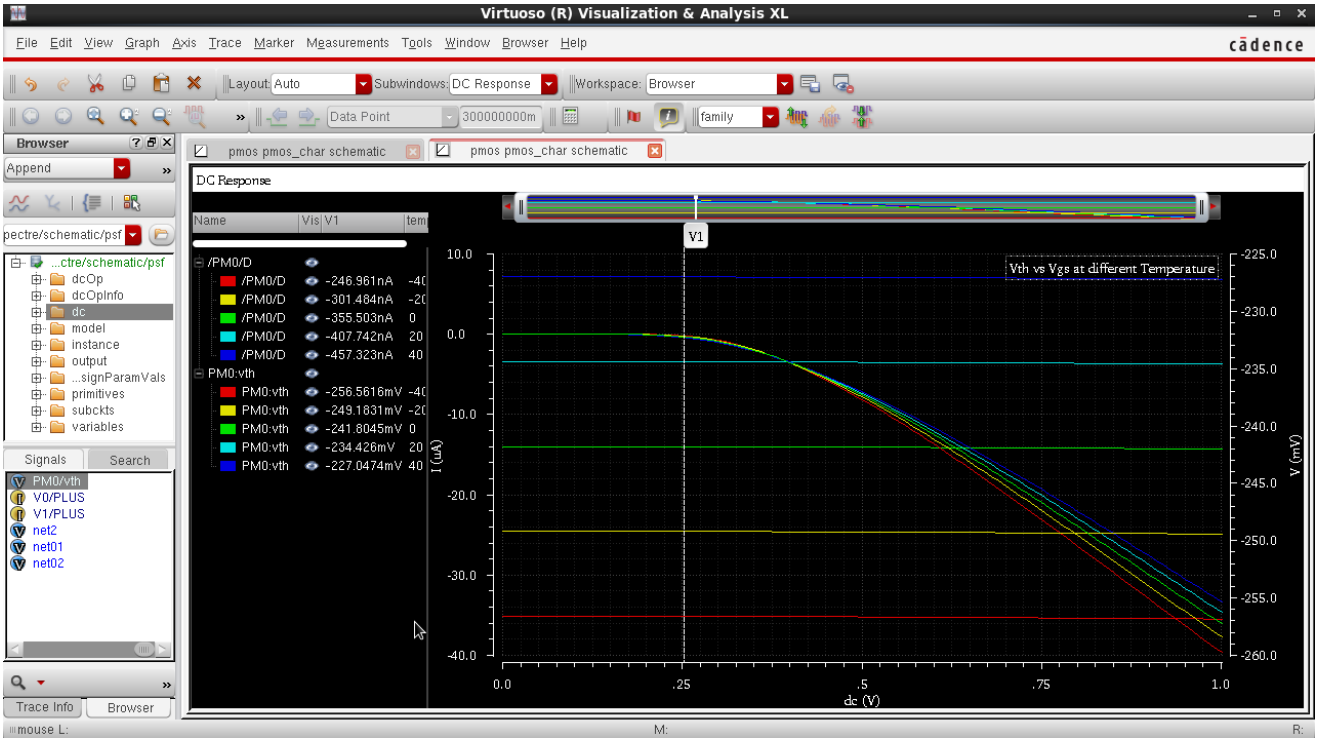
  - At low drain current, the transistor provides high transconductance efficiency, resulting in a high g_m/I_d.
  - As | V_{gs} |increases, the drain current grows faster than g_m, so the transconductance efficiency (g_m/I_d) decreases.

- **Region v/s V_{gs}:**



  - At low V_{gs}, the PMOS operates in Region 3 (saturation) because $|V_{gs}| > |V_{th}|$, resulting in strong channel formation.
  - Around $V_{GS} \approx 0.23$ V, the region changes from saturation (3) to linear (2) as the drain-source voltage no longer satisfies the saturation condition.

- **V_{th} v/s V_{gs}:**



  - As temperature increases from −40°C to 40°C, the PMOS threshold voltage magnitude decreases (becomes less negative), causing the device to turn ON at a lower gate voltage, while the drain current slightly decreases due to reduced carrier mobility.